

2-Fredholm formula ①

$$\Pi(\bar{u}, p) = \frac{1}{2} \int_{\Omega} u_{i,j} (2\mu) u_{i,j} dV + \int_{\Omega} p (u_{i,i} - \frac{p}{2\lambda}) dV - \int_{\Omega} u_i b_i dV - \int_{\partial\Omega_h} u_i t_i ds$$

Derive weak form — $\delta_{\bar{u}, p} \Pi = 0$

Consider variations:

$$\bar{u}_\epsilon = \bar{u} + \epsilon \bar{w}$$

$$p_\epsilon = p + \epsilon w^p$$

$$\left. \frac{d}{d\epsilon} \left[\Pi(\bar{u} + \epsilon \bar{w}, p + \epsilon w^p) \right] \right|_{\epsilon=0}$$

$$\begin{aligned} \Pi(\bar{u} + \epsilon \bar{w}, p + \epsilon w^p) &= \frac{1}{2} \int_{\Omega} (u_{i,j} + \epsilon w_{i,j}) (2\mu) (u_{i,j} + \epsilon w_{i,j}) dV \\ &+ \int_{\Omega} (p + \epsilon w^p) \left(u_{i,i} + \epsilon w_{i,i} - \frac{p + \epsilon w^p}{2\lambda} \right) dV \\ &- \int_{\Omega} (u_i + \epsilon w_i) b_i dV - \int_{\partial\Omega_h} (u_i + \epsilon w_i) t_i ds \end{aligned}$$

~~$\delta_{\bar{u}, p} \Pi = \frac{d\Pi}{d\epsilon}$~~

$$\begin{aligned} \left. \frac{d\Pi(\bar{u} + \epsilon \bar{w}, p + \epsilon w^p)}{d\epsilon} \right|_{\epsilon=0} &= \frac{1}{2} \int_{\Omega} w_{i,j} 2\mu u_{i,j} dV + \frac{1}{2} \int_{\Omega} u_{i,j} 2\mu w_{i,j} dV \\ &+ \int_{\Omega} w^p \left(u_{i,i} - \frac{p}{2\lambda} \right) dV + \int_{\Omega} p \left(w_{i,i} - \frac{w^p}{2\lambda} \right) dV \\ &- \int_{\Omega} w_i b_i dV - \int_{\partial\Omega_h} w_i t_i ds \end{aligned}$$

(2.1)

$$\int_{\Omega} \omega^p u_{i,i} dV = \sum_e \int_{\Omega_e} \left(\sum_{A=1}^{Npe} N^A \beta^A \right) \left(\sum_{B=1}^{Nne} N_{,i}^B d_i^B \right) dV$$

$$= \sum_e \sum_A \sum_B \beta^A \left(\int_{\Omega_e} N^A N_{,i}^B dV \right) \cdot d_i^B$$

$$= \sum_e \sum_A \sum_B \beta^A \left(\int_{\Omega^y} N^A(z) \cdot N_{,i}^B z_{,i} \cdot \det(J) dV^y \right) d_i^B$$

$(Npe \times 1)$ G_e^T $(Nne \times 1)$
 $(Npe \times Nne)$

$$= \sum_e \beta_e^T G_e^T d_e = \beta^T G^T d$$

(2.2)

$$\int_{\Omega} \omega_{i,i} p dV = \sum_e \int_{\Omega_e} \left(\sum_{A=1}^{Nne} N_{,i}^A c_i^A \right) \left(\sum_{B=1}^{Npe} N^B \alpha^B \right) dV$$

$$= \sum_e \sum_A \sum_B c_i^A \left(\int_{\Omega_e} N_{,i}^A N^B dV \right) \alpha^B$$

$$= \sum_e \sum_A \sum_B c_i^A \left(\int_{\Omega^y} N_{,i}^A z_{,i} N^B \det(J) dV^y \right) \alpha^B$$

$(Nne \times 1)$ G_e $(Npe \times 1)$
 $(Nne \times Npe)$

$$= \sum_e c_e^T G_e \alpha_e$$

$$= c^T G \alpha$$

$$\begin{aligned}
 \textcircled{3} \quad - \int_{\Omega} \omega^P \cdot \frac{P}{\lambda} dV &= - \sum_e \int_{\Omega_e} \left(\frac{\sum_A^{Npe} N^A \beta^A}{\lambda} \right) \left(\sum_B^{Npe} N^B \alpha^B \right) dV \\
 &= - \sum_e \sum_{A,B}^{Npe} \beta^A \left(\int_{\Omega_e} \frac{N^A N^B}{\lambda} dV \right) \alpha^B \\
 &= - \sum_e \sum_{A,B}^{Npe} \beta^A \underbrace{\left(\int_{\Omega_e} \frac{N^A(z) N^B(z)}{\lambda} \det(J) dV^z \right)}_{\substack{Me \\ \rightarrow Npe \times Npe}} \underbrace{\alpha^B}_{(Npe \times 1)} \\
 &= \sum_e \beta_e M_e \alpha_e = \beta M \alpha
 \end{aligned}$$

Now, the entire MV system we have:

$$C^T K d + \beta^T G^T d + C^T G \alpha + \beta M \alpha =$$

RHS formed from the body force & Neumann BC integrals

$$\begin{matrix} \leftarrow C^T & \beta^T \rightarrow \end{matrix} \begin{bmatrix} K & G \\ G^T & M \end{bmatrix} \begin{bmatrix} d \\ \alpha \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

How to solve this?

for nearly incompressible material, $(\nu \rightarrow 0.5)$; $M \rightarrow 0$: M^{-1} is not defined

$$\rightarrow K d + G \alpha = F_1 \Rightarrow \boxed{d = K^{-1} (F_1 - G \alpha)} \quad *$$

$$\rightarrow G^T d = F_2$$

$$\Rightarrow G^T K^{-1} [F_1 - G \alpha] = F_2$$

$$\Rightarrow \boxed{G^T K^{-1} G \alpha = G^T K^{-1} F_1 - F_2}$$

$\hookrightarrow$ solve for α

$\downarrow$
substitute in * to get d